

Límites

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e$$

$$\lim_{x \rightarrow 0} \frac{x}{\sin x} = 1$$

$$\lim_{x \rightarrow \infty} \left(1 + \frac{k}{x}\right)^x = e^k$$

$$\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$$

$$\lim_{x \rightarrow \infty} \left(1 + \frac{k}{x+a}\right)^{x+a} = e^k$$

Derivadas

$$f(x) = k \Rightarrow f'(x) = 0$$

$$f(x) = x \Rightarrow f'(x) = 1$$

$$f(x) = cx \Rightarrow f'(x) = c$$

$$f(x) = x^n \Rightarrow f'(x) = n \cdot x^{n-1}$$

$$f(x) = \ln x \Rightarrow f'(x) = \frac{1}{x}$$

$$f(x) = \sin x \Rightarrow f'(x) = \cos x$$

$$f(x) = \cos x \Rightarrow f'(x) = -\sin x$$

$$f(x) = e^x \Rightarrow f'(x) = e^x$$

$$f(x) = \log_a x \Rightarrow f'(x) = \frac{1}{x \ln a}$$

$$f(x) = \tan x \Rightarrow f'(x) = \sec^2 x$$

$$f(x) = \cot x \Rightarrow f'(x) = -\csc^2 x$$

$$f(x) = \sec x \Rightarrow f'(x) = \sec x \tan x$$

$$f(x) = \csc x \Rightarrow f'(x) = -\csc x \cot x$$

$$f(x) = a^x \Rightarrow f'(x) = a^x \ln a$$

$$f(x) = \tan^{-1} x \Rightarrow f'(x) = \frac{1}{1+x^2}$$

$$f(x) = \sin^{-1} x \Rightarrow f'(x) = \frac{1}{\sqrt{1-x^2}}$$

$$f(x) = \cos^{-1} x \Rightarrow f'(x) = -\frac{1}{\sqrt{1-x^2}}$$

$$f(x) = \sinh x \Rightarrow f'(x) = \cosh x$$

$$f(x) = \cosh x \Rightarrow f'(x) = \sinh x$$

$$f(x) = \tanh^{-1} x \Rightarrow f'(x) = \frac{1}{1-x^2}$$

$$f(x) = \sinh^{-1} x \Rightarrow f'(x) = \frac{1}{\sqrt{1+x^2}}$$

$$f(x) = \cosh^{-1} x \Rightarrow f'(x) = \frac{1}{\sqrt{x^2-1}}$$

Reglas de Derivación

$$y = f(x)g(x) \Rightarrow y' = f'(x)g(x) + f(x)g'(x)$$

$$y = \frac{f(x)}{g(x)} \Rightarrow y' = \frac{f'(x)g(x) - f(x)g'(x)}{g^2(x)}$$

$$h(x) = f(g(x)) \Rightarrow h'(x) = f'(g(x)) \cdot g'(x)$$

Integrales

$$\int dx = \int 1 dx = x + C$$

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

$$\int \frac{1}{x} dx = \ln |x| + C$$

$$\int e^x dx = e^x + C$$

$$\int \sin x dx = -\cos x + C$$

$$\int \cos x dx = \sin x + C$$

$$\int a^x dx = \frac{a^x}{\ln a} + C$$

$$\int \sinh x dx = \cosh x + C$$

$$\int \cosh x dx = \sinh x + C$$

$$\int \sec^2 x dx = \tan x + C$$

$$\int \csc^2 x dx = -\cot x + C$$

$$\int \sec x \tan x dx = \sec x + C$$

$$\int \csc x \cot x dx = -\csc x + C$$

$$\int \frac{1}{1+x^2} dx = \tan^{-1} x + C$$

$$\int \frac{1}{\sqrt{1-x^2}} dx = \sin^{-1} x + C$$

$$\int \frac{1}{1-x^2} dx = \tanh^{-1} x + C$$

$$\int \frac{1}{\sqrt{1+x^2}} dx = \sinh^{-1} x + C$$

$$\int \frac{1}{\sqrt{x^2-1}} dx = \cosh^{-1} x + C$$

Identidades Trigonométricas

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$\sin^2 x + \cos^2 x = 1$$

$$\tan^2 x + 1 = \sec^2 x$$

$$\cosh^2 x - \sinh^2 x = 1$$

$$\sin(2x) = 2 \sin x \cos x$$

$$\tan x = \frac{\sin x}{\cos x}$$

$$\cot x = \frac{\cos x}{\sin x}$$

$$\sec x = \frac{1}{\cos x}$$

$$\csc x = \frac{1}{\sin x}$$

$$\sin^2 x = \frac{1 - \cos(2x)}{2}$$

$$\cos^2 x = \frac{1 + \cos(2x)}{2}$$

$$\sinh x = \frac{e^x - e^{-x}}{2}$$

$$\cosh x = \frac{e^x + e^{-x}}{2}$$

Logaritmos

$$\log_a 1 = 0$$

$$\log_a a = 1$$

$$a^{\log_a x} = x$$

$$\log_b n = x \Leftrightarrow b^x = n$$

$$\log_a (x/y) = \log_a x - \log_a y$$

$$\log_a (x \cdot y) = \log_a x + \log_a y$$

$$\log_b (a^n) = n \cdot \log_b a$$

$$\log_{b^n} (a^n) = \log_b a$$

$$\log_b a = \frac{1}{\log_a b}$$

$$\log_b a = \frac{\log_c a}{\log_c b}$$

Valor Absoluto

$$|a| = |-a|$$

$$|ab| = |a||b|$$

$$\left| \frac{a}{b} \right| = \frac{|a|}{|b|}$$

$$|x + y| \leq |x| + |y|$$

Álgebra Lineal

Teorema de Laplace:

$$\det(B) = \sum_{j=1}^n (-1)^{i+j} B_{i,j} M_{i,j}$$

Donde $M_{i,j}$ es el determinante de la submatriz obtenida al remover la i -ésima fila y la j -ésima columna de B

$$D(x) = d(x) c(x) + R(x)$$

$$\frac{D(x)}{d(x)} = c(x) + \frac{R(x)}{d(x)}$$

Cónicas

Para saber el centro (h, k) sustituir x con $(x - h)$ e y con $(y - k)$

Circunferencia

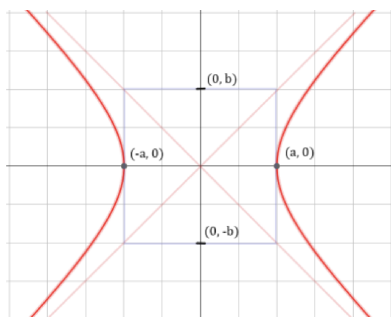
$$x^2 + y^2 = r^2$$

Elipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

Hipérbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$



$$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$$

